

QUESTION 1

Code:

```
# Data

data <- c(31, 35, 28, 42, 37, 33,
         45, 29, 39, 34, 41, 36)
# Sample mean
observed_mean <- mean(data)

cat("Sample Mean =", observed_mean, "\n")
# Bootstrap resampling
set.seed(123)

B <- 5000
bootstrap_means <- replicate(

  B,

  mean(sample(data, size = length(data), replace = TRUE))

)
# 95% Bootstrap Confidence Interval
CI <- quantile(bootstrap_means, c(0.025, 0.975))
cat("95% Bootstrap Confidence Interval:\n")

print(CI)

# Bootstrap distribution
hist(

  bootstrap_means,
```

```

breaks = 30,

main = "Bootstrap Distribution of Sample Mean",

xlab = "Bootstrap Mean",

col = "lightblue",

border = "white"

)
# Mark observed mean

abline(

  v = observed_mean,

  col = "red",

  lwd = 2

)

legend(

  "topright",

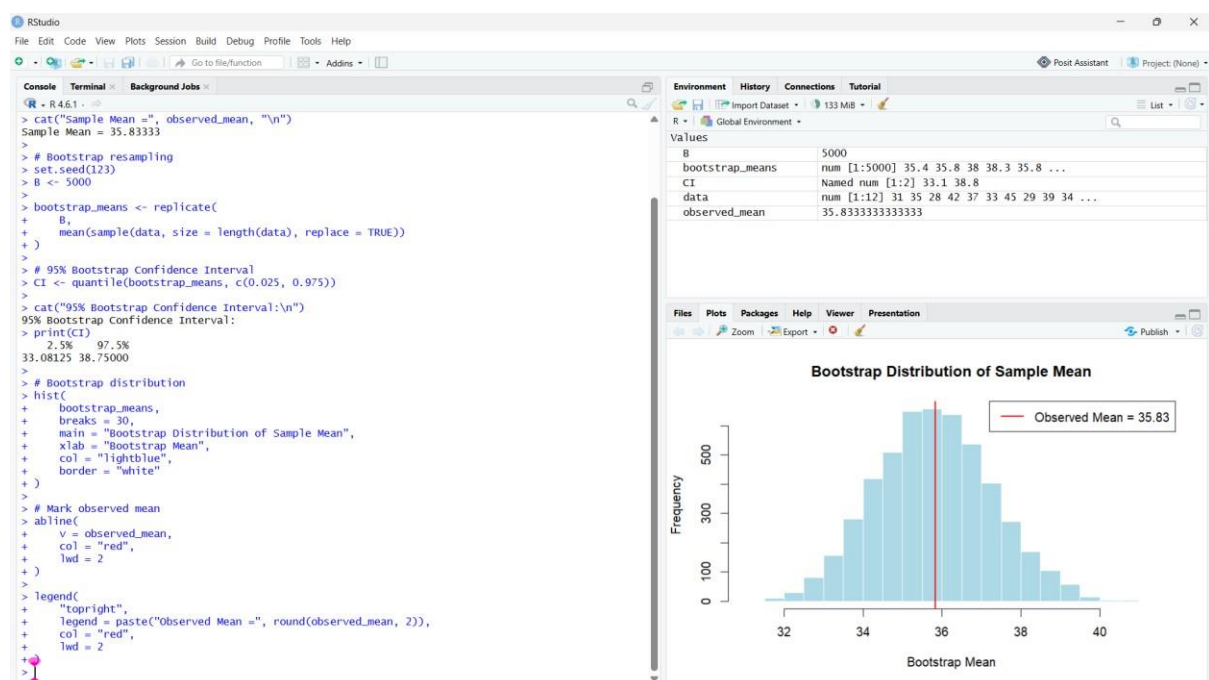
  legend = paste("Observed Mean =", round(observed_mean, 2)),

  col = "red",

  lwd = 2

)

```



Interpretation

- **Sample mean = 35.83 minutes**
- The bootstrap procedure generates **5,000 resamples** from the original 12 observations.
- The **95% bootstrap confidence interval** gives a range of plausible values for the population mean delivery time.
- The histogram shows the distribution of bootstrap sample means, with the **observed sample mean marked by the red line**.

QUESTION 2

Code:

```
library(ggplot2)
# Data

data <- data.frame(
  Advertising = c(10,12,15,18,20,23,25,28,30,33,36,40),
  Sales = c(48,51,53,57,60,61,66,69,72,75,79,84)
)

# Linear regression model

model <- lm(Sales ~ Advertising, data = data)
# Model summary

summary(model)
```

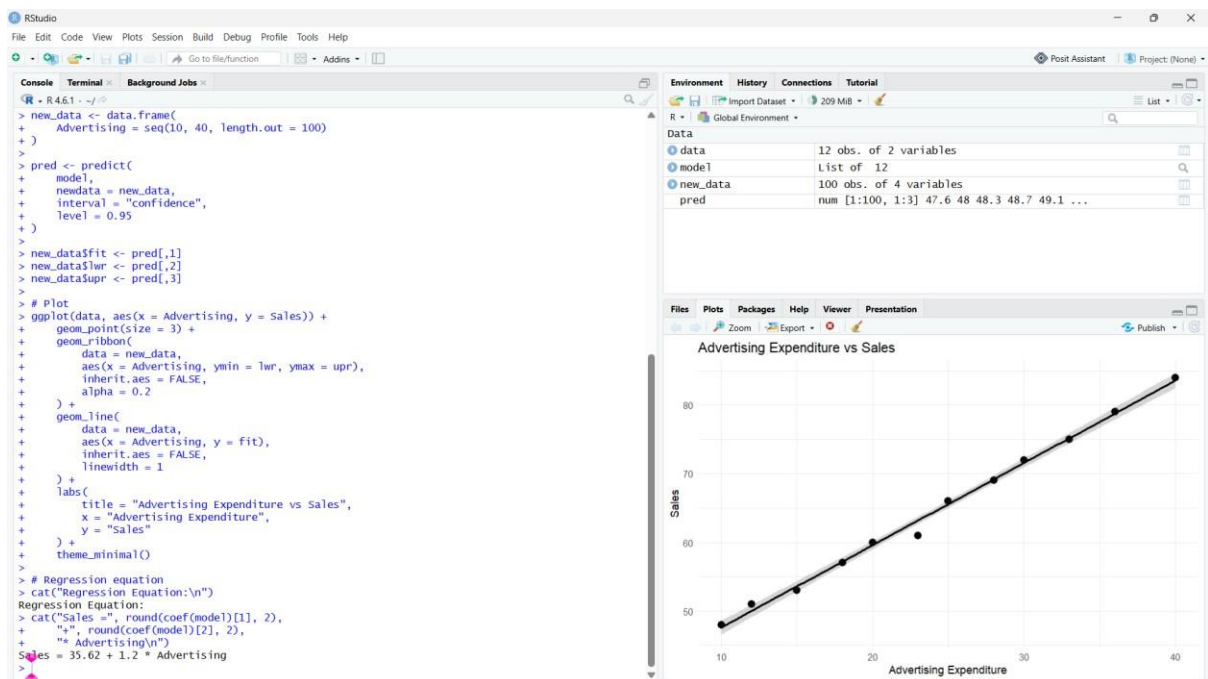
```
# Prediction data
new_data <- data.frame(
  Advertising = seq(10, 40, length.out = 100)
)
pred <- predict(
  model,
  newdata = new_data,
  interval = "confidence",
  level = 0.95
)

new_data$fit <- pred[,1]
new_data$lwr <- pred[,2]
new_data$upr <- pred[,3]
# Plot
ggplot(data, aes(x = Advertising, y = Sales)) +
  geom_point(size = 3) +
  geom_ribbon(
    data = new_data,
    aes(x = Advertising, ymin = lwr, ymax = upr),
    inherit.aes = FALSE,
    alpha = 0.2
  ) +
  geom_line(
    data = new_data,
    aes(x = Advertising, y = fit),
    inherit.aes = FALSE,
    linewidth = 1
  )
```

```

) +
labs(
  title = "Advertising Expenditure vs Sales",
  x = "Advertising Expenditure",
  y = "Sales"
) +
theme_minimal()
# Regression equation
cat("Regression Equation:\n")
cat("Sales =", round(coef(model)[1], 2),
    "+", round(coef(model)[2], 2),
    "* Advertising\n")

```



Interpretation

- The regression line shows a **positive relationship** between advertising expenditure and sales.
- As advertising expenditure increases, **sales also increase**.
- The **95% confidence band** represents uncertainty around the estimated mean sales.
- The band is generally **narrower near the center of the advertising range** and becomes wider toward the lower and upper extremes.

QUESTION 3

Code:

```
library(ggplot2)
# Data
data <- data.frame(
  Hours = c(2,3,3,4,4,4,5,5,5,5,
            6,6,6,6,6,7,7,7,7,7,
            8,8,8,8,8,9,9,9,10,10),
  Scores = c(45,52,52,58,60,60,64,66,66,66,
            70,72,72,72,72,76,78,78,80,80,
            84,84,85,85,85,88,90,90,94,94)
)
# 1. Original scatterplot
```

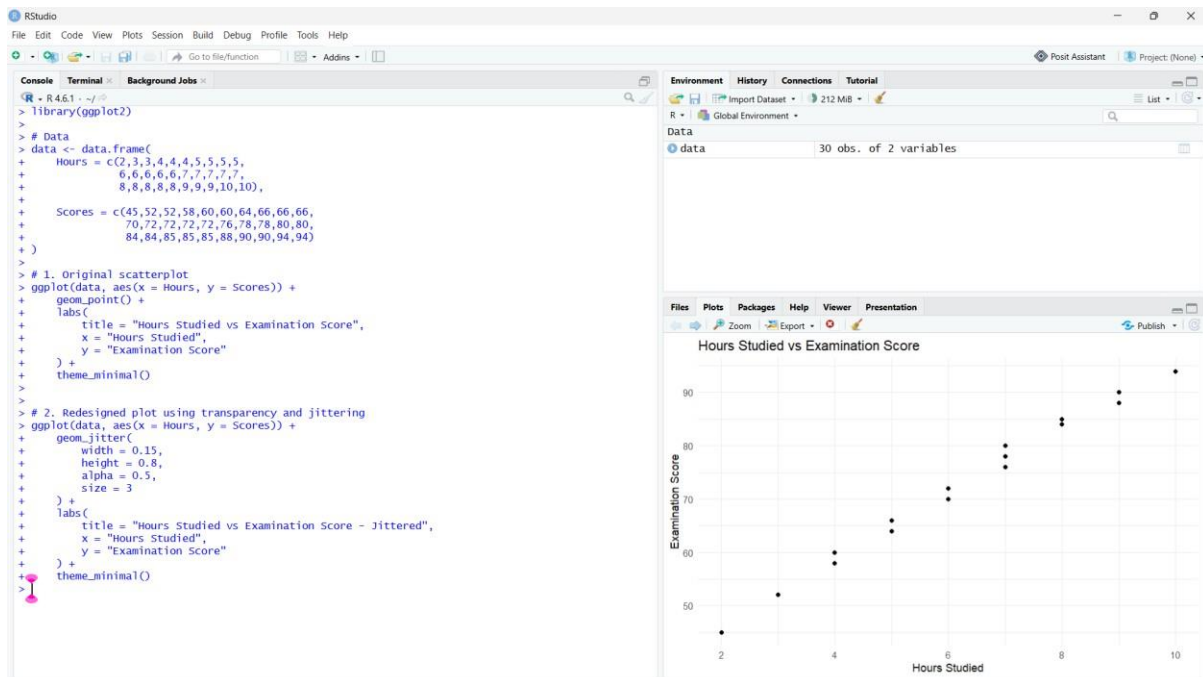
```
ggplot(data, aes(x = Hours, y = Scores)) +  
geom_point() +  
  labs(  
    title = "Hours Studied vs Examination Score",  
    x = "Hours Studied",  
    y = "Examination Score"  
  ) +  
  theme_minimal()  
# 2. Redesigned plot using transparency and jittering  
ggplot(data, aes(x = Hours, y = Scores)) +  
geom_jitter(  
  width = 0.15,  
  height = 0.8,  
  alpha = 0.5,  
  size = 3  
)+  
  labs(  
    title = "Hours Studied vs Examination Score - Jittered",  
    x = "Hours Studied",  
    y = "Examination Score"  
  ) +  
  theme_minimal()
```

Interpretation

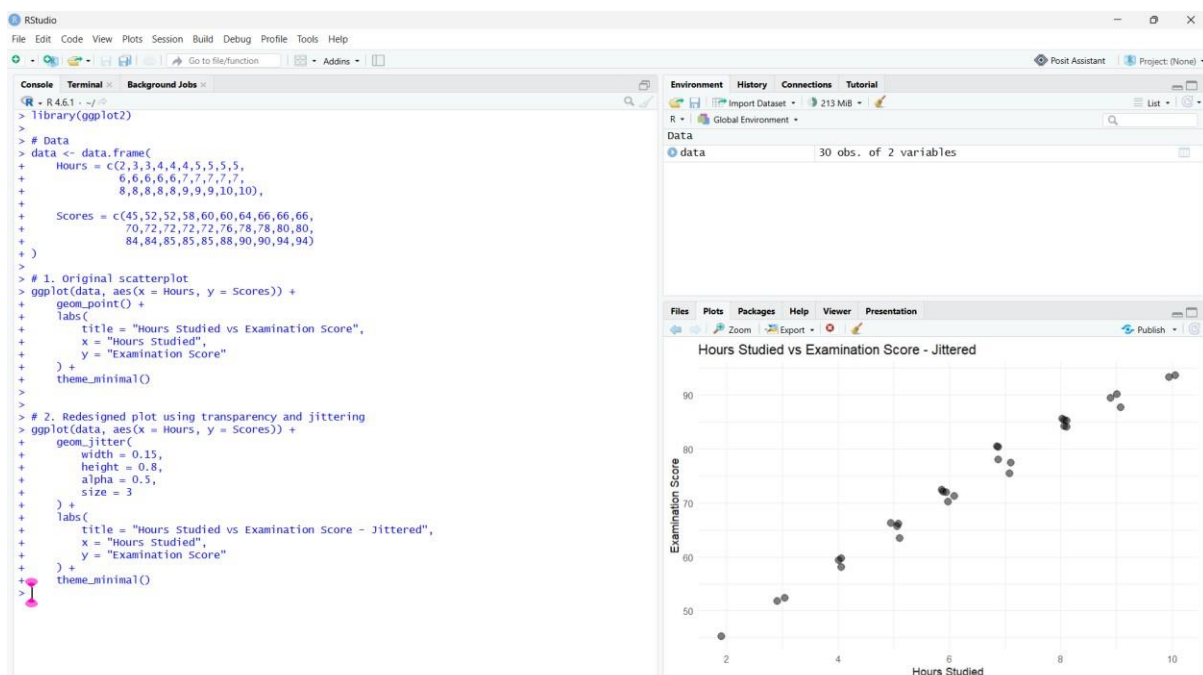
- In the **original scatterplot**, identical observations overlap, making some points appear as a single point.
- This is called **overplotting** and can hide how many students have the same Hours–Score combination.

- The **jittered plot** slightly moves overlapping points so individual observations become visible.
- The **transparency (alpha)** also helps identify areas where many observations overlap.
- The redesigned plot therefore gives a clearer view of the **concentration and distribution of students' scores**.

Output:



Output 2:



QUESTION 4

Code:

```
library(ggplot2)
# Data

data <- data.frame(

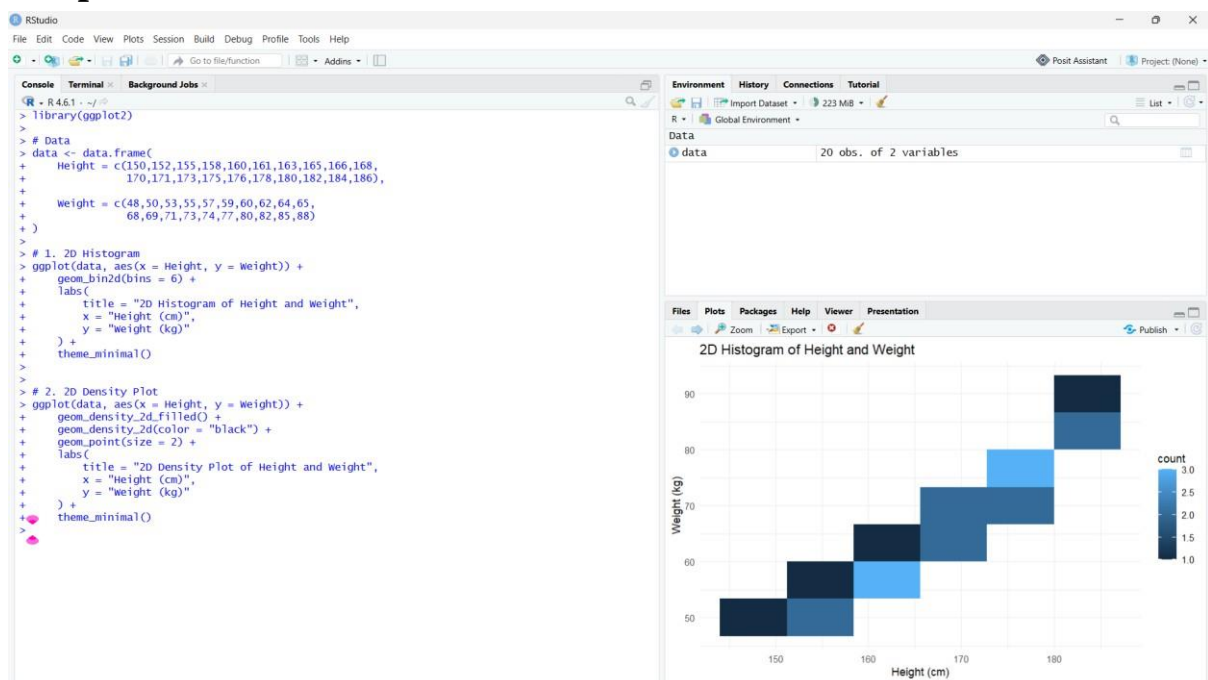
  Height = c(150,152,155,158,160,161,163,165,166,168,
            170,171,173,175,176,178,180,182,184,186),

  Weight = c(48,50,53,55,57,59,60,62,64,65,
            68,69,71,73,74,77,80,82,85,88)

)
# 1. 2D Histogram
ggplot(data, aes(x = Height, y = Weight)) +
  geom_bin2d(bins = 6) +
  labs(
    title = "2D Histogram of Height and Weight",
    x = "Height (cm)",
    y = "Weight (kg)"
  ) +
  theme_minimal()
# 2. 2D Density Plot
ggplot(data, aes(x = Height, y = Weight)) +
  geom_density_2d_filled() +
  geom_density_2d(color = "black") +
  geom_point(size = 2) +
```

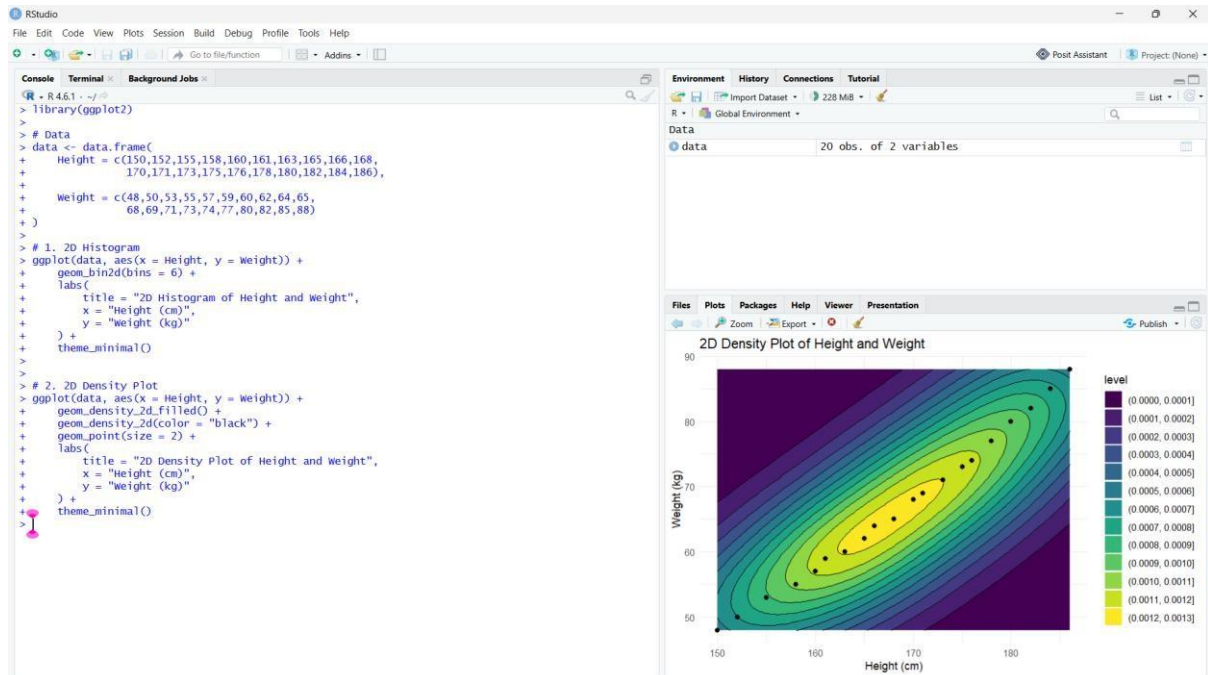
```
labs(
  title = "2D Density Plot of Height and Weight",
  x = "Height (cm)",
  y = "Weight (kg)"
) +
theme_minimal()
```

Output:



Interpretation

- The **2D histogram** divides the height-weight space into rectangular bins and shows the number of observations in each region.
- The **2D density plot** estimates the concentration smoothly and uses contours to identify regions with higher observation density.
- The observations show a clear **positive relationship**: as height increases, weight generally increases.



QUESTION 5

Code:

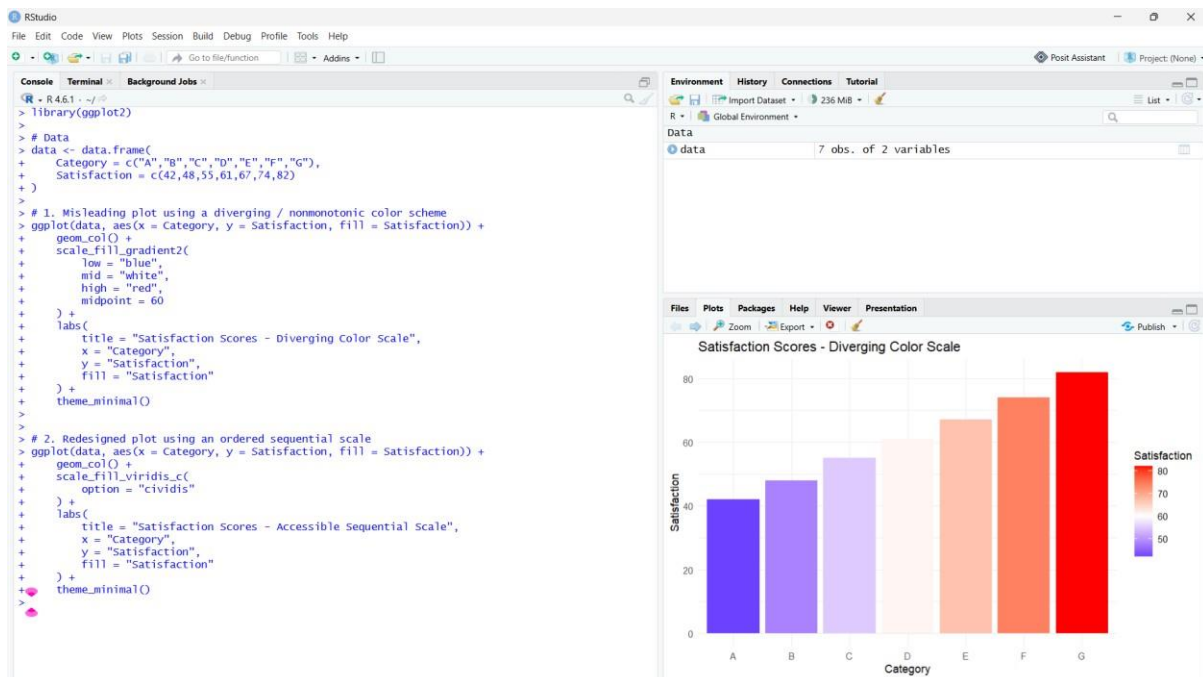
```
library(ggplot2)
# Data
data <- data.frame(
  Category = c("A", "B", "C", "D", "E", "F", "G"),
  Satisfaction = c(42, 48, 55, 61, 67, 74, 82)
)
# 1. Misleading plot using a diverging / nonmonotonic color scheme
ggplot(data, aes(x = Category, y = Satisfaction, fill = Satisfaction)) +
  geom_col() +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
```

```

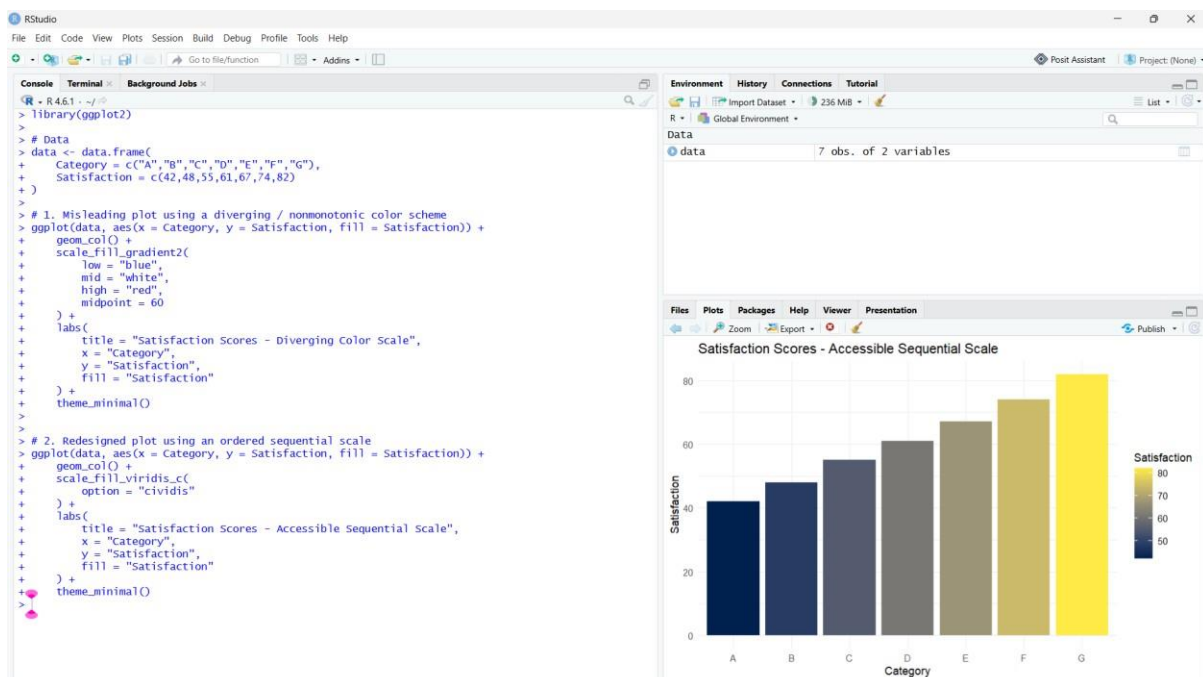
    high = "red",
    midpoint = 60
) +
labs(
  title = "Satisfaction Scores - Diverging Color Scale",
  x = "Category",
  y = "Satisfaction",
  fill = "Satisfaction"
) +
theme_minimal()
# 2. Redesigned plot using an ordered sequential scale
ggplot(data, aes(x = Category, y = Satisfaction, fill = Satisfaction)) +
  geom_col() +
  scale_fill_viridis_c(
    option = "cividis"
) +
labs(
  title = "Satisfaction Scores - Accessible Sequential Scale",
  x = "Category",
  y = "Satisfaction",
  fill = "Satisfaction"
) +
theme_minimal()

```

Output:



Output 2:



misleading design choices

1 – Bootstrapping and Confidence Intervals

- Insufficient number of bootstrap resamples..
- Poor axis labels or scaling can make the bootstrap distribution difficult to interpret.

2 – Uncertainty in Trend Estimation

- Missing or unclear confidence band.
- Inappropriate axis scaling can exaggerate the trend.
- Confusing confidence interval with prediction interval.
- Extending the regression line far beyond the observed data can be misleading.

3 – Overplotting, Jittering and Transparency

- Overlapping points can hide the number of observations.
- Fully opaque points make dense regions appear to contain fewer observations.
- Excessive jitter can make points appear at values that were not actually observed.
- Too much transparency can make dense areas difficult to distinguish.

4 – 2D Histogram and Density Estimation

- Too few or too many bins can distort the apparent concentration.
- Unequal or inappropriate axis scales can exaggerate relationships.
- Excessive smoothing can hide important patterns.
- Density contours may suggest concentrations that are not strongly supported by a small dataset.